

Tutorial - 4

1. Draw the signal flow graph corresponding to the control system block diagram provided in Fig. 1. Find the overall closed-loop transfer function (from $R(s)$ to $Y(s)$) using Mason's Gain Formula.

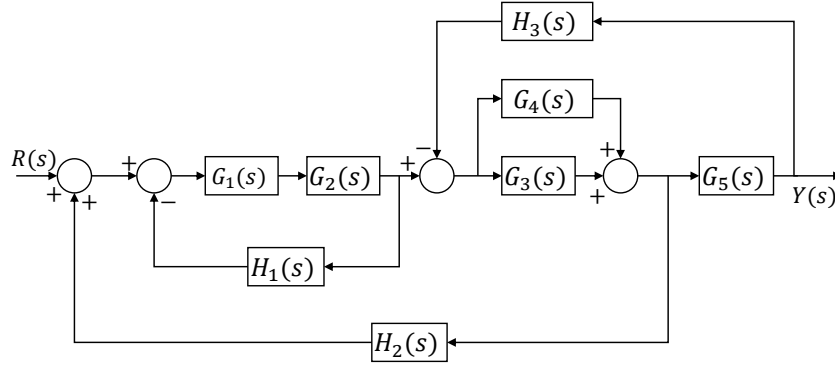


Figure 1

Answer:

$$\frac{Y(s)}{R(s)} = \frac{G_1(s)G_2(s)G_5(s)(G_3(s) + G_4(s))}{(1 + G_1(s)G_2(s)H_1(s))\{1 + (G_3(s) + G_4(s))G_5(s)H_3(s)\} - G_1(s)G_2(s)(G_3(s) + G_4(s))H_2(s)}.$$

2. The damping ratio (ζ) and natural frequency (ω_n) of a standard second order system are 0.6 and 2 rad/s, respectively. Find the peak time, rise time, settling time and peak overshoot for the unit step response of the system. Also obtain the steady state error due to the unit-step input for the unity feedback closed-loop system.

Answer: Peak time $t_p = 1.95\text{sec}$

Settling time $t_s = 3.3\text{sec}$

Rise time $t_r = 1.38\text{sec}$

Peak overshoot = 9.48%

Steady state error $e_{ss} = 0$.

3. Consider the control systems shown in Fig. 2, where τ is a positive real constant. Find the steady-state error due to a unit step reference if the controller gain is tuned to $K_p = 10$.

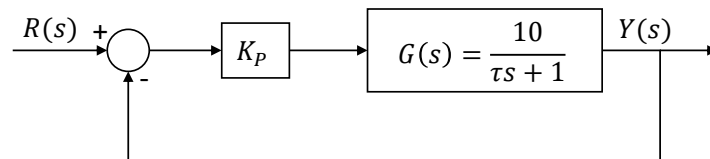


Figure 2

Answer: $e_{ss} = 0.0099$.

4. Consider the system shown in Fig. 3, where $w(t)$ is a unit-step disturbance. Calculate the range of K such that the steady-state value of $y(t)$ due to $w(t)$ is less than one percent of $w(t)$.

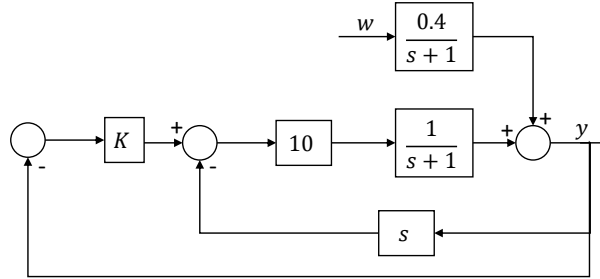


Figure 3

Answer: $K > 3.9$.

5. Consider that the open-loop transfer function of a unity negative feedback control system is $\frac{s+6}{Ks^2}$. Find the value of K to achieve a damping ratio of 0.5.

Answer: $K = \frac{1}{6}$.

6. The open-loop transfer function of a unity negative feedback control system is given by

$$G(s) = \frac{K}{s(s+2)}.$$

Find the value of K for the peak overshoot of the closed-loop system to a unit step input to be 10%.

Answer: $K = 2.87$.